

EXACT FINITE CERTIFICATION OF THE SPARSE-COMPLEMENT TWO-EIGENVALUE CONJECTURE AT ORDER NINE

ZIJIAN ZENG

ABSTRACT. For a graph G , let $q(G)$ be the minimum number of distinct eigenvalues of a real symmetric matrix whose off-diagonal support is exactly $E(G)$. A conjecture originating with Barrett, Fallat, Furst, Nasserar, Rooney, and Tait asserts that $e(\overline{G}) \leq |V(G)| - 3$ implies $q(G) = 2$. We give a computer-assisted exact certification of the first order left open by the cited literature: every graph G on nine vertices with $e(\overline{G}) \leq 6$ has $q(G) = 2$. The certification reconstructs all 108 isomorphism types. In a disjoint routing, 72 cases use the published bipartite-complement theorem, 24 use the published balanced connected-join theorem, and the remaining 12 have explicit rank-three algebraic projection certificates encoded by rational directions and positive rational weights. An independent verifier rebuilds the graph list by connected-component multisets, checks its counts by Burnside's lemma, and recomputes all decisive identities with rational arithmetic. The same verification independently reproduces the known orders seven and eight. No claim is made for order ten or for the full conjecture.

1. INTRODUCTION

For a simple graph G of order n , let $S(G)$ be the set of real symmetric $n \times n$ matrices $A = (a_{ij})$ such that, for $i \neq j$,

$$a_{ij} \neq 0 \iff ij \in E(G).$$

The diagonal is unrestricted. If $q(A)$ is the number of distinct eigenvalues of A , set

$$q(G) = \min\{q(A) : A \in S(G)\}.$$

If G' is obtained from G by relabeling vertices with a permutation matrix Π , then $A \mapsto \Pi A \Pi^T$ is a spectrum-preserving bijection from $S(G)$ to $S(G')$. Thus it is enough to certify one representative of each graph-isomorphism class. Barrett et al. restated the conjecture that

$$(1) \quad e(\overline{G}) \leq n - 3 \implies q(G) = 2$$

for every $n \geq 3$ [1, Conjecture 1.1]. The conjecture had first appeared publicly in Fallat–Mojallal, explicitly based on a 2022 personal communication from those six authors [2, Conjecture 1 and reference 37]. Fallat–Mojallal

proved the cases $n = 7, 8$ [2, Theorems 22–23], and Barrett et al. proved (1) whenever $\overline{G}$ is bipartite [1, Theorem 3.7].

The purpose of this paper is the following finite result.

Theorem 1.1. *If G is a simple graph on nine vertices and $e(\overline{G}) \leq 6$, then $q(G) = 2$.*

The proof is exact but computer-assisted. Floating-point optimization was used only to discover short certificate data; neither the theorem nor its verifier uses a numerical tolerance. The certificate also reconstructs the orders seven and eight, rather than taking their published verification as a computational input.

2. EXACT REDUCTIONS

We first record the normalization and the short certificate lemma used by the verification.

Lemma 2.1 (orthogonal normalization). *Let G be a nonempty graph. Then $q(G) = 2$ if and only if there is a real symmetric $Q \in S(G)$ such that $Q^2 = I$.*

Proof. If $A \in S(G)$ has exactly the eigenvalues $\lambda \neq \mu$, then

$$Q = \frac{2A - (\lambda + \mu)I}{\lambda - \mu}$$

is symmetric, has the same off-diagonal support, and satisfies $Q^2 = I$. Conversely, a symmetric involution has spectrum contained in $\{-1, 1\}$. It cannot be scalar because G is nonempty, so both eigenvalues occur. $\square$

Lemma 2.2 (weighted Parseval certificate). *Let $H = \overline{G}$ have vertex set $\{1, \dots, n\}$. Suppose that $0 < r < n$, that $w_i \in \mathbb{Q}^r$, and that $x_i \in \mathbb{Q}_{>0}$ satisfy*

$$(2) \quad w_i^\top w_j = 0 \iff ij \in E(H) \quad (i \neq j),$$

$$(3) \quad \sum_{i=1}^n x_i w_i w_i^\top = I_r.$$

Then $q(G) = 2$. More precisely, these data determine an algebraic symmetric involution in $S(G)$.

Proof. Let row i of V be $\sqrt{x_i} w_i^\top$. Equation (3) says $V^\top V = I_r$; hence $P = VV^\top$ is a rank- r orthogonal projection. Positivity of the weights and (2) give, for $i \neq j$,

$$P_{ij} = \sqrt{x_i x_j} w_i^\top w_j = 0 \iff ij \in E(H).$$

Thus $Q = I - 2P$ is a symmetric involution whose off-diagonal support is exactly $E(G)$. Its eigenvalues are -1 with multiplicity r and $+1$ with multiplicity $n - r$; hence both occur when $0 < r < n$. Lemma 2.1 finishes the proof. $\square$

We also use the following immediate graph reduction. Suppose the connected components of H are divided into two nonempty unions H_L, H_R . There are no H -edges between the two vertex sets, and therefore

$$(4) \quad G = \overline{H_L} \vee \overline{H_R}.$$

If the two factors in (4) are connected and their orders differ by at most two, then $q(G) = 2$ by the connected case of [3, Theorem 3.4].

3. EXHAUSTIVE EXACT CERTIFICATION

3.1. Completeness of the finite list. The certification uses two independent exact decompositions. First, every graph is uniquely, up to the order of its factors, a disjoint union of connected components. A nontrivial connected graph with at most six edges has at most seven vertices, since it contains a spanning tree. We therefore take the connected isomorphism types through order seven from the graph atlas [5], form every multiset within the relevant vertex and edge budgets, and fill unused vertices with isolated vertices. Uniqueness of connected components prevents two different multisets from representing isomorphic graphs.

Second, the verifier checks the number of resulting types separately for each edge count by Burnside's lemma. For each cycle type of S_n , it forms the induced permutation on the unordered pairs of vertices. If the pair-orbit lengths are $d_1, \dots, d_s$, the edge enumerator of the fixed graphs is

$$\prod_{j=1}^s (1 + z^{d_j}).$$

The coefficient sums, weighted by conjugacy-class sizes and divided by $n!$, agree with the component enumeration. The totals are 19, 44, and 108 for $n = 7, 8, 9$, respectively. The discovery enumeration using **geng** [4] is retained as a further cross-check but is not imported by the exact verifier.

3.2. Disjoint routing. Each graph $H = \overline{G}$ is passed through the following routes in order.

- (1) If H is bipartite, apply [1, Theorem 3.7].
- (2) Otherwise, test every bipartition of the connected components of H . If (4) has connected factors whose orders differ by at most two, apply [3, Theorem 3.4].
- (3) For the residual isomorphism types, verify a certificate satisfying Lemma 2.2 with $r = 3$.

The exact counts are as follows.

For transparency, the canonical graph6 identifiers of the complements $H = \overline{G}$ in the order-nine exact-frame cases are

```
H????CN  H????C^  H????KN  H????KZ
H??G?CV  H??G?Cr  H??G?CN  H???OK[
H???_SF  H???GoM  H???OKT  H???GKX.
```

TABLE 1. Disjoint route counts.

n	all types	bipartite	balanced join	exact frame
7	19	16	2	1
8	44	33	10	1
9	108	72	24	12

The baseline residual identifiers are $F??Gw$ at order seven and $G???G\{$ at order eight. These strings are identifiers only; the verifier parses their graphs and matches isomorphism types, rather than trusting the strings as mathematical claims.

One of the 12 order-nine exact-frame complements has five edges and the other 11 have six. This route residue is determined only after applying the two theorems above; it is intentionally not the same as the smaller literature-only frontier obtained by also using every reduction in the cited paper.

The serialized certificate records every coordinate of every w_i and every x_i as a reduced rational. The verifier checks positivity, all $\binom{n}{2}$ zero/nonzero decisions in (2), and all entries of (3) using exact fractions. Unknown schema fields, noncanonical rationals, incomplete lists, and endpoint changes are rejected.

Proposition 3.1. *The certificate routes every isomorphism class in Table 1; all 14 residual classes satisfy Lemma 2.2 with $r = 3$.*

Proof. This is a finite exact calculation by the verifier described above. Its input SHA-256 digest is

d78b49e23eaebba29d0ab70661df9688a09baabce7e0333040b43186a808fd5d.

The verifier prints its own code digest, the input digest, every route count, and the word **VERIFIED**. Eighteen mutation tests cover missing or altered witnesses, incorrect counts, noncanonical graph6 and decimal encodings, duplicate or unknown keys, nonstandard JSON constants, nulls, and type confusions between integers, floats, and Booleans; all are rejected. The complete rational data and verification commands are included with the source archive. $\square$

Proof of Theorem 1.1. Take $H = \overline{G}$. Proposition 3.1 exhausts every isomorphism type with $|V(H)| = 9$ and $e(H) \leq 6$. The first route gives $q(G) = 2$ by the published bipartite-complement theorem, the second by the published balanced connected-join theorem and (4), and the third by Lemma 2.2. Finally, G has at least $\binom{9}{2} - 6 = 30$ edges, so $q(G) \neq 1$. Therefore $q(G) = 2$ in every case. $\square$

Remark 3.2 (baseline reproduction). The same argument and verifier certify all $n = 7, 8$ cases with edge bounds four and five. Thus the previously

published last baseline is reconstructed from the serialized data rather than assumed by the new order-nine enumeration.

4. SCOPE, REPRODUCIBILITY, AND DISCLOSURE

The theorem is finite. It does not prove the conjecture for $n \geq 10$, nor does it classify minimum multiplicity bipartitions. Its mathematical trust boundary consists of the two cited published theorems, the elementary lemmas in Section 2, and the finite exact verification in Section 3.

A two-pass source and novelty search, last repeated on August 29, 2026 after the manuscript endpoint had frozen, found no prior public full resolution of the order-nine endpoint or exact certificate for it. This negative novelty statement is limited to the databases, queries, and date recorded with the source archive; it is not a claim about unpublished or unindexed work.

From the project root, the decisive commands are

```
python verification/verify_n9_certificate.py \
    certificates/n9_exact_frames.json
python verification/test_fail_closed.py
```

The recorded verification environment uses Python 3.13.5 and NetworkX 3.4.2. The verifier requires no optimizer or computer algebra system. The discovery record separately pins NetworkX 3.4.2, NumPy 2.2.3, SciPy 1.15.3, SymPy 1.13.3, and nauty/Traces **geng** 2.9.3. Discovery used deterministic seeds and floating-point Stiefel and direct-involution searches, followed by rational reconstruction. Those floating-point results are retained as falsification evidence only and are neither read nor trusted by the verifier.

Generative AI assisted literature search, route design, code drafting, symbolic reconstruction, adversarial test design, and exposition. The mathematical conclusion is bound to the exact certificate and independently executable verifier. No Lean, Coq, Isabelle, or other proof assistant was used; the result is therefore not claimed to be proof-assistant formalized.

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INSTITUTE OF COMPUTER SCIENCE AND DIGITAL INNOVATION, UCSI UNIVERSITY,
KUALA LUMPUR, 56000, MALAYSIA

Email address: zijianzeng@foxmail.com

Email address: 1002266693@ucsiuniversity.edu.my